

Homework — 1.3.4 Web Technologies

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.3.4 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. A web page uses HTML, CSS and JavaScript together. A developer wants every paragraph styled with the same blue text, and wants one specific warning box styled differently.

(a) State which **CSS selector type** would be used to style **all** paragraphs the same way. **[1]** (AO1)

(b) State which **CSS selector type** (`.class` or `#id`) is most appropriate for styling the **single** warning box, and justify your choice. **[2]** (AO2)

2. Explain why **CSS** is used to control presentation rather than placing styling directly in the HTML of every page of a large website. **[2]** (AO2)

3. Describe the role of a **web crawler** in how a search engine builds its index. **[2]** (AO1)

4. The **PageRank** algorithm ranks pages by importance.

(a) Explain how an incoming link from a **high-ranking** page affects a page's rank differently from a link from a **low-ranking** page. **[2]** (AO1)

(b) State what is meant by saying PageRank is calculated **iteratively** until it **converges**. [2] (AO1)

5. A web form lets a user choose a username. The browser checks instantly that the box is not empty (client-side), and the server then checks that the username is not already taken in the database (server-side).

(a) Give **one** benefit of doing the "not empty" check **client-side**. [1] (AO2)

(b) Explain why the "username already taken" check **must** be done server-side. [2] (AO2)

Part B — Retrieval: keep it fresh (~6 marks)

6. (1.3.3) State the difference between a **switch** and a **router**, referring to the type of address each uses. [2] (AO1)

7. (1.3.1) Explain **one** reason why **symmetric** encryption has a key distribution problem that **asymmetric** encryption avoids. [2] (AO1)

8. (1.4.1) Convert the denary number 45 into 8-bit binary. Show your working. [2] (AO2)
